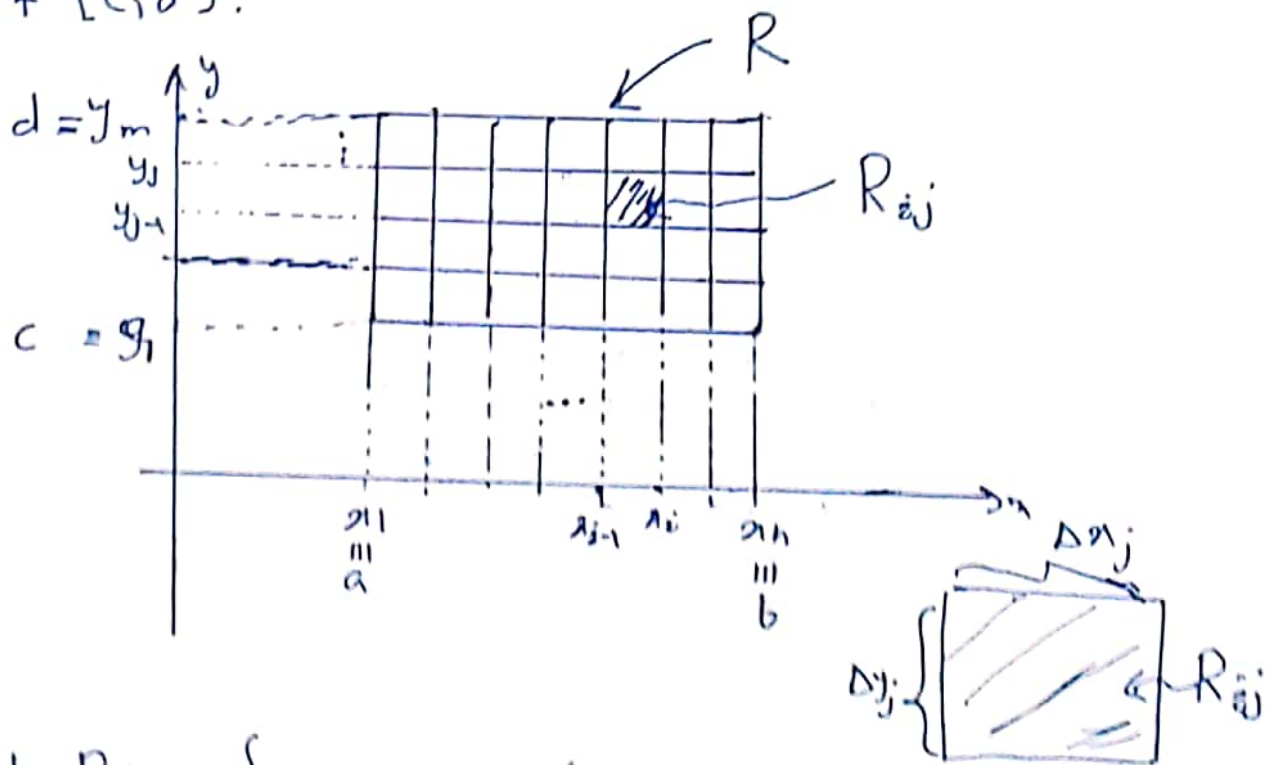


— Multiple Integrals —

Let $R = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, c \leq y \leq d\}$.

Let $f: R \rightarrow \mathbb{R}$ be a bounded function.

Consider the partition P of R into $m \times n$ rectangles created by the partitions $\{x_1, x_2, \dots, x_n\}$ of $[a, b]$ and $\{y_1, y_2, \dots, y_m\}$ of $[c, d]$.

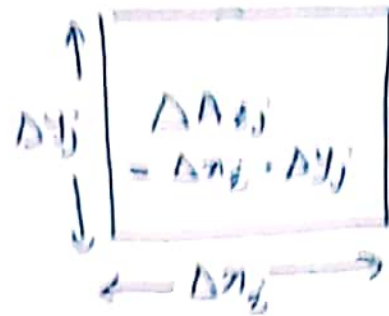


Let $R_{ij} = \{(x, y) \in \mathbb{R}^2 \mid x_{j-1} \leq x \leq x_j, y_{j-1} \leq y \leq y_j\}$

for $i = 1, 2, \dots, n$ and

$j = 1, 2, \dots, m$

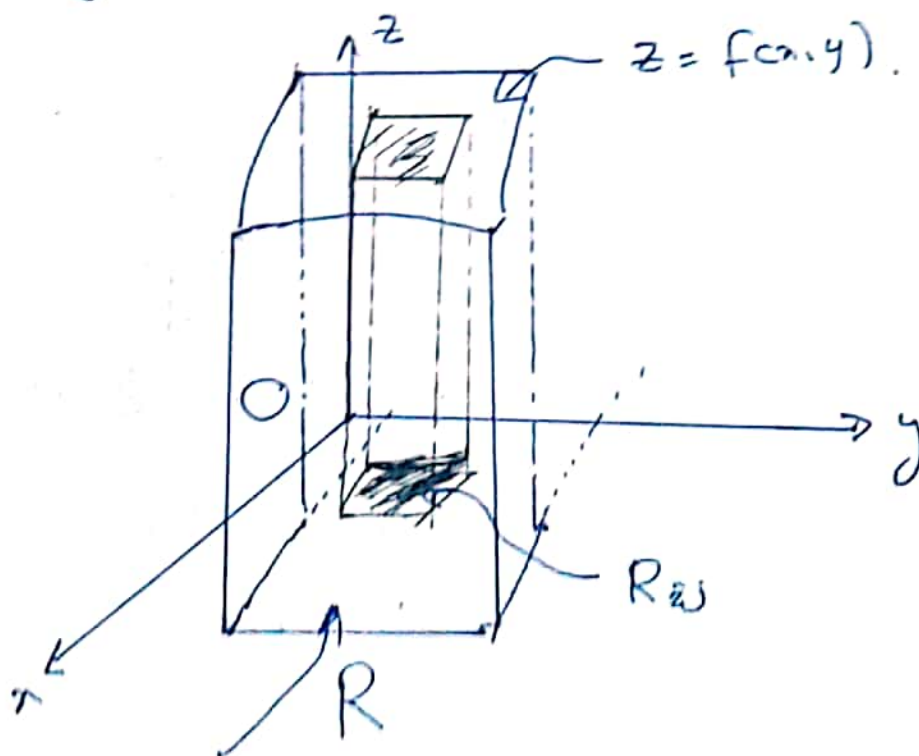
Let $\Delta x_i = x_i - x_{i-1}$ for $i = 1, 2, \dots, n$ and
 $\Delta y_j = y_j - y_{j-1}$ for $j = 1, 2, \dots, m$.



The area of $R_{ij} = \Delta A_{ij} = \Delta x_i \times \Delta y_j$
 for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

Let $m_{ij} = \inf \{ f(x, y) / (x, y) \in R_{ij} \}$

$M_{ij} = \sup \{ f(x, y) / (x, y) \in R_{ij} \}$



Now form the upper sum

$$U(f, P) = \sum_{i=1}^n \sum_{j=1}^m M_{ij} \Delta A_{ij} \text{ and}$$

the lower sum

$$L(f, P) = \sum_{i=1}^n \sum_{j=1}^m m_{ij} \Delta A_{ij}.$$

Let $U(f) = \inf \{ U(f, P) / P \text{ is a partition of } R \}$
and $L(f) = \sup \{ L(f, P) / P \text{ is a partition of } R \}$.

If $U(f) = L(f)$, then f is said to be Riemann integrable over R .

This common value is denoted by

$$\iint_R f(x, y) dA \text{ or } \iint_R f(x, y) dx dy.$$

Note:

If $f(x, y) \geq 0$ for $(x, y) \in R$ and f is Riemann integrable over R , then the volume of the solid that lies above R and

Now form the upper sum

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$$\text{and} \quad L(f) = \sup \{ L(f, P) / P \text{ is a partition of } R \}.$$

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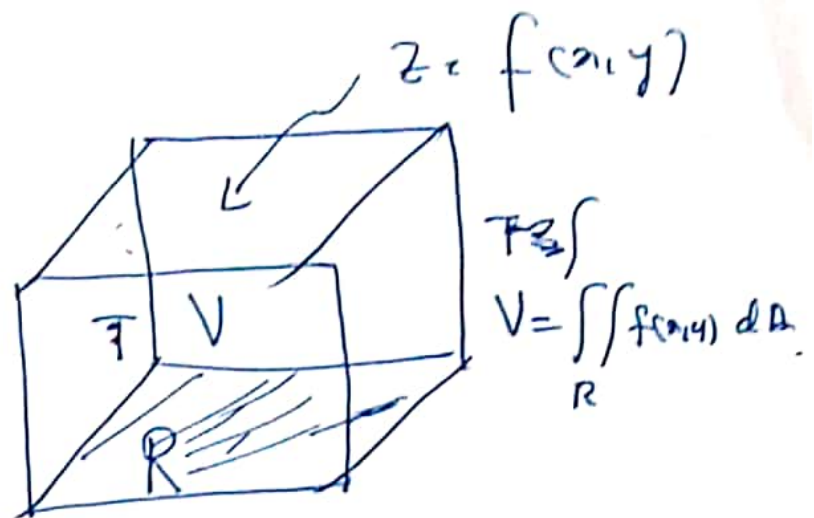
This common value is denoted by

$$\iint_R f(x, y) dA \quad \text{or} \quad \int \int_R f(x, y) dx dy.$$

Note:

If $f(x, y) \geq 0$ for $(x, y) \in R$ and f is Riemann integrable over R , then the volume of the solid that lies above R and

Under the surface $z = f(x, y)$ is defined⁴
 by $\iint_R f(x, y) dA$.

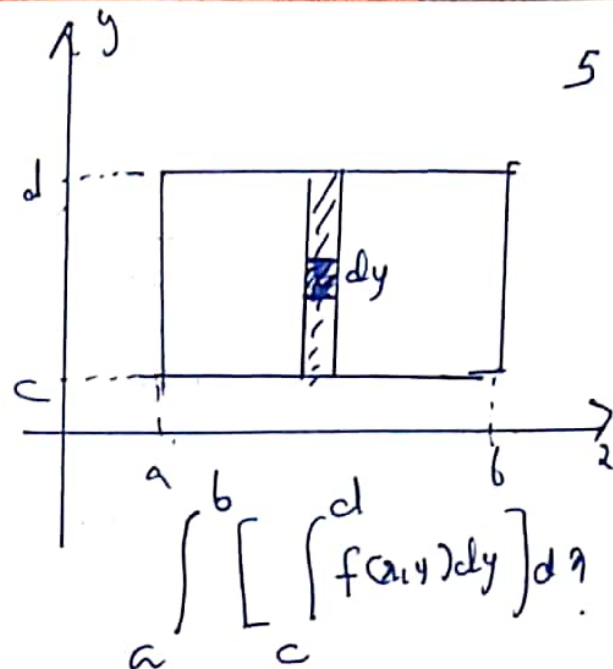
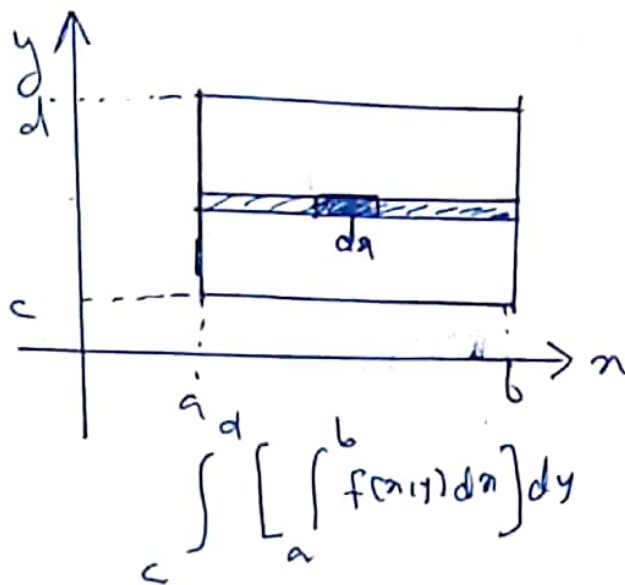


Theorem 1

Let f be Riemann integrable over the rectangle $R = \{(x, y) \in \mathbb{R}^2 / a \leq x \leq b, c \leq y \leq d\}$
 $= [a, b] \times [c, d]$.

Then,

$$\begin{aligned} \iint_R f(x, y) dA &= \int_c^d \left[\int_a^b f(x, y) dx \right] dy \\ &= \int_a^b \left[\int_c^d f(x, y) dy \right] dx. \end{aligned}$$

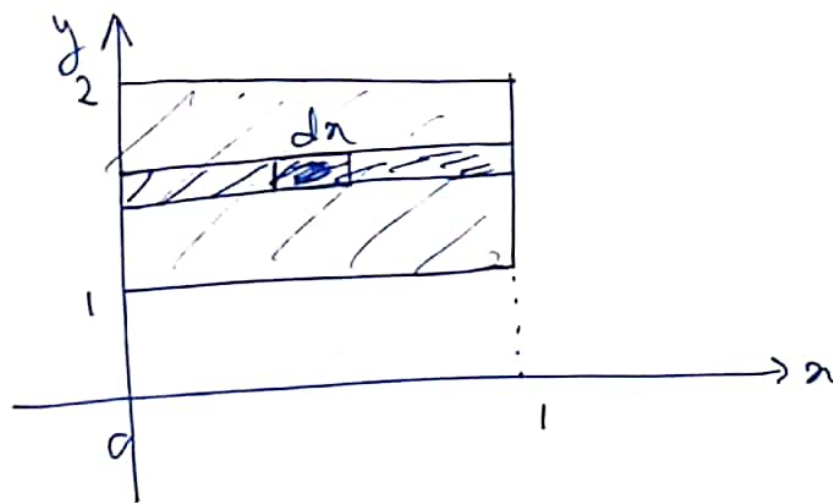


Example

Evaluate $\iint_R (xy - x^3) dA$, where

$$R = \left\{ (x,y) \in \mathbb{R}^2 / 0 \leq x \leq 1, 1 \leq y \leq 2 \right\}.$$

Let $f(x,y) = xy - x^3$.



By Theorem 1, we have

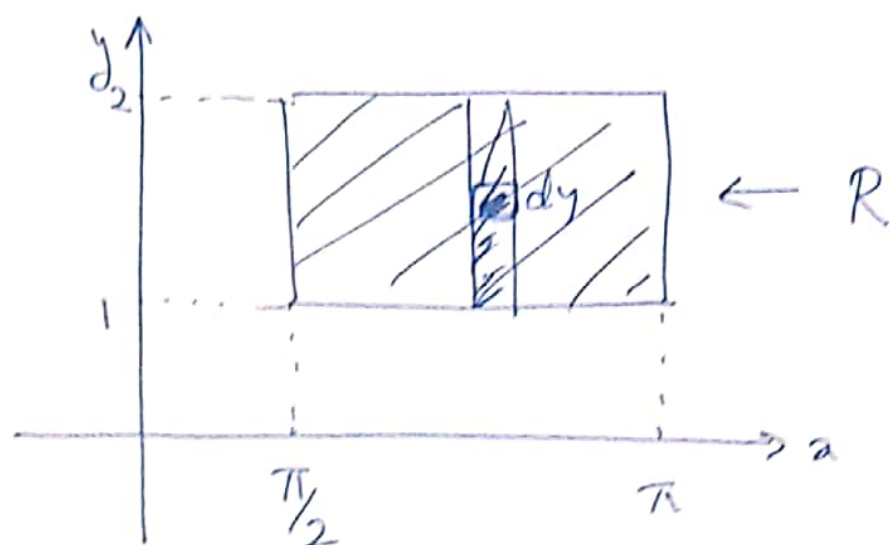
6

$$\begin{aligned}\iint_R f(x, y) dA &= \int_{y=1}^2 \left[\int_{x=0}^1 f(x, y) dx \right] dy \\&= \int_{y=1}^2 \left[\int_{x=0}^1 (xy - x^3) dx \right] dy \\&= \int_{y=1}^2 \left[y \cdot \frac{x^2}{2} - \frac{x^4}{4} \right]_{x=0}^1 dy \\&= \int_{y=1}^2 \left(\frac{y}{2} - \frac{1}{4} - 0 \right) dy \\&= \left[\frac{y^2}{4} - \frac{y}{4} \right]_{y=1}^2 \\&= \frac{1}{4} \left[(2^2 - 2) - (1^2 - 1) \right] \\&= \frac{1}{2} \quad \square\end{aligned}$$

Example

Evaluate $\iint_R x \sin(xy) dx dy$, where

$$R = \left\{ (x, y) \in \mathbb{R}^2 / \frac{\pi}{2} \leq x \leq \pi, 1 \leq y \leq 2 \right\}.$$



$$\iint_R x \sin(xy) dx dy = \int_{x=\pi/2}^{\pi} \left[\int_{y=1}^2 x \sin(xy) dy \right] dx$$

$$= \int_{x=\pi/2}^{\pi} \left[-\cos(xy) \right]_{y=1}^2 dx$$

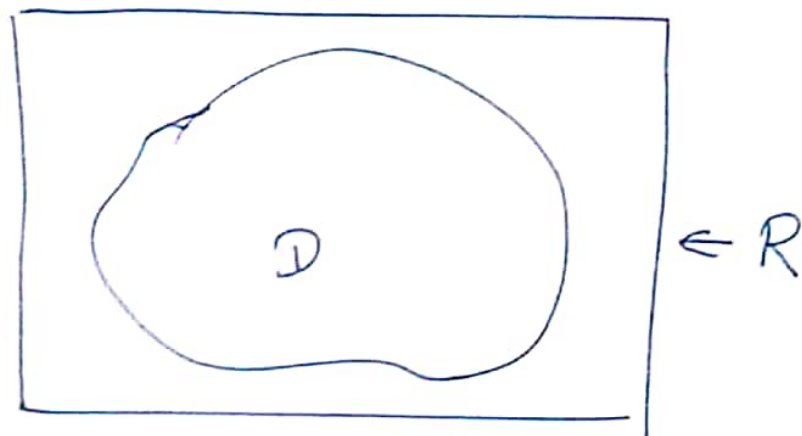
$$= \int_{x=\pi/2}^{\pi} (\cos x - \cos 2x) dx$$

$$= \left[\sin x - \frac{\sin 2x}{2} \right]_{x=\pi/2}^{\pi}$$

$$= \left(\sin \pi - \frac{\sin 2\pi}{2} \right) - \left(\sin \frac{\pi}{2} - \frac{\sin \pi}{2} \right)$$

$$= -1 \quad \square$$

— Double integration over general region —



Let D be a bounded region.

Then there exists a rectangle R containing D .

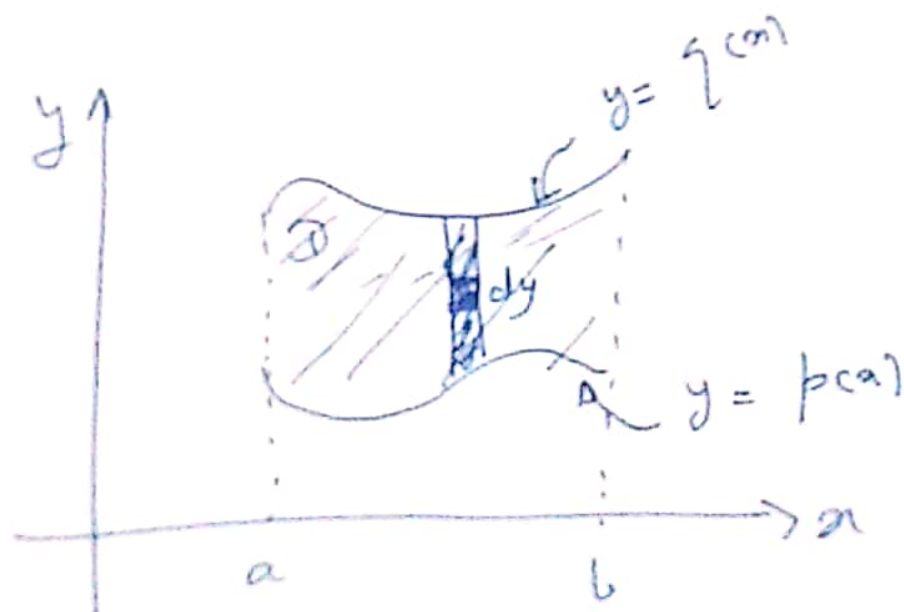
Define a new function g on R by

$$g(x,y) = \begin{cases} f(x,y) & \text{if } (x,y) \in D \\ 0 & \text{if } (x,y) \in D' \end{cases}$$

If g is integrable over R , then we say that f is integrable over D and we define the double integral of f over D by

$$\begin{aligned} \iint_D f(x,y) dA &= \iint_R g(x,y) dA \\ &= \iint_R g(x,y) dx dy \end{aligned}$$

Type 1



$$D = \left\{ (x, y) \in \mathbb{R}^2 / p(x) \leq y \leq q(x), a \leq x \leq b \right\} \quad \text{Id}$$

We may prove that

$$\iint_D f(x, y) dA = \int_{x=a}^b \left[\int_{y=p(x)}^{q(x)} f(x, y) dy \right] dx.$$

Example

Let D be the region bounded by the parabolas: $y = 2x^2$ and $y = 1 + x^2$.

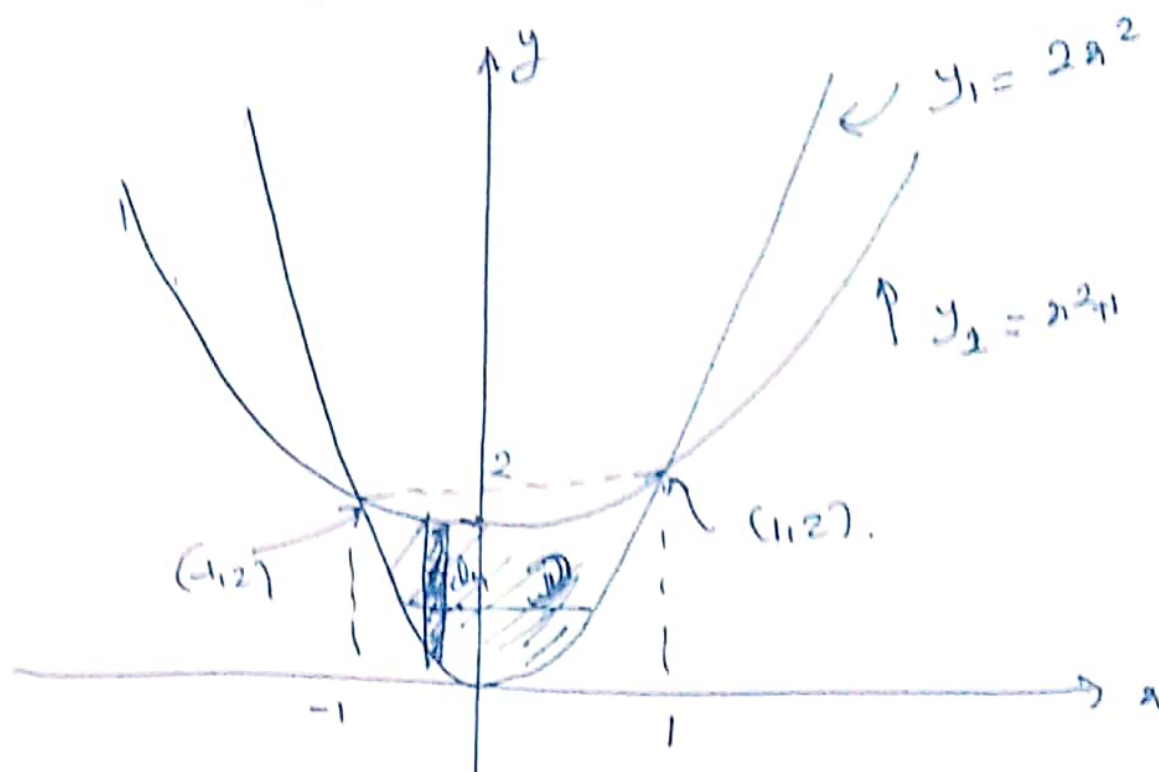
Evaluate $\iint_D (3x + 6y) dA$.

Let $y_1 = p(x) = 2x^2$ and $y_2 = q(x) = 1 + x^2$.

The points of the intersection of y_1 and y_2 are given by

$$y_1 = y_2 \Leftrightarrow 2x^2 = 1 + x^2 \Leftrightarrow x = \pm 1$$

The intersection points are : $(1, 2)$, $(-1, 2)$.



This is Type 1.

$$\text{So, } \iint_D (3x + 6y) dA = \int_{x=-1}^1 \left[\int_{y=2x^2}^{1+x^2} (3x + 6y) dy \right] dx$$

$$= \int_{x=-1}^1 \left[3xy + 3y^2 \right]_{y=2x^2}^{1+x^2} dx$$

$$= \int_{x=-1}^1 3 \left[\left\{ x(1+x^2) + (1+x^2)^2 \right\} - \left\{ x \cdot 2x^2 + (2x^2)^2 \right\} \right] dx$$

$$= \int_{x=-1}^1 3 (-3x^4 - x^3 + 2x^2 + x + 1) dx$$

$$= 3 \left[-\frac{32^4}{4} \right]$$

$$= 3 \left[-\frac{3x^5}{5} - \frac{x^4}{4} + \frac{2x^3}{3} - \frac{x^2}{2} + x \right]_{x=-1}^1$$

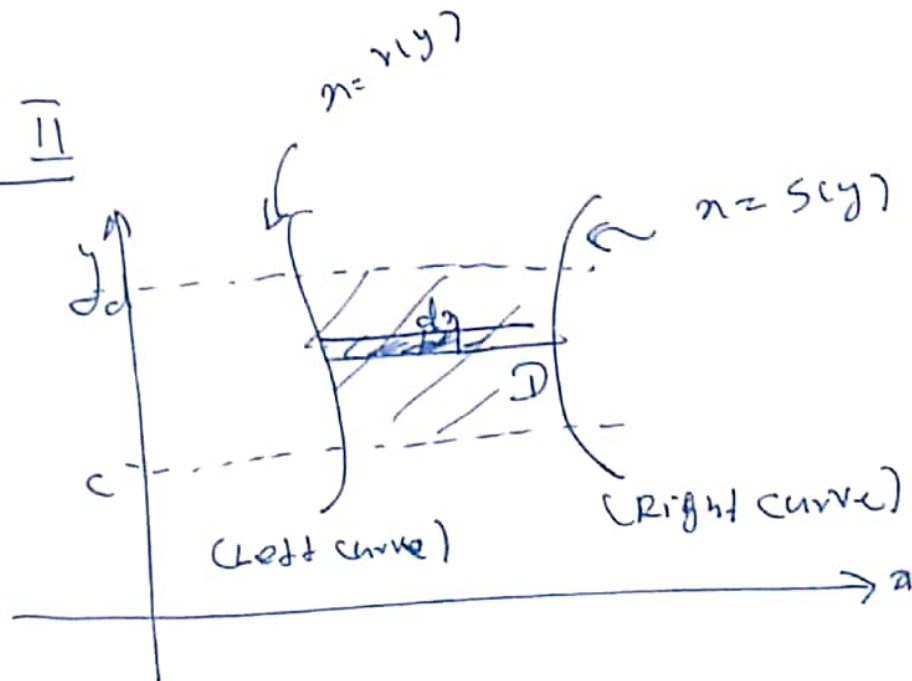
$$= 3 \left[\left(-\frac{3}{5} - \frac{1}{4} + \frac{2}{3} - \frac{1}{2} + 1 \right) - \left(\frac{3}{5} - \frac{1}{4} - \frac{2}{3} + \frac{1}{2} - 1 \right) \right]$$

$$= 3 \left[-\frac{6}{5} + \frac{4}{3} + 2 \right]$$

$$= 3 \cdot \left(\frac{32}{15} \right)$$

$$= \frac{32}{5} \quad \square$$

Type II



$r(y) \leq s(y)$ for all $y \in [c, d]$ and r and s are continuous on $[c, d]$.

$$D = \left\{ (x, y) \in \mathbb{R}^2 \mid v(y) \leq x \leq s(y), c \leq y \leq d \right\}$$

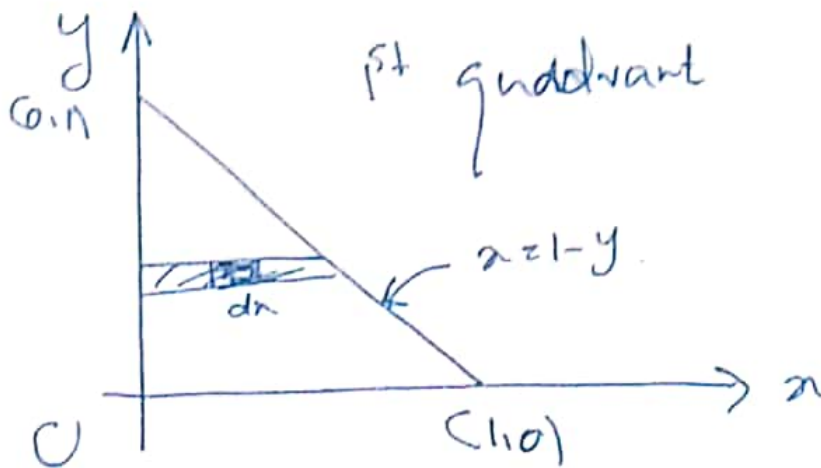
As in type I case, we have:

if f is continuous on D , then

$$\iint_D f(x, y) dA = \int_{y=c}^d \left[\int_{x=v(y)}^{s(y)} f(x, y) dx \right] dy$$

Example

Evaluate $\iint_D e^{x+y} dx dy$, where D is the region in the first quadrant bounded by the co-ordinate axes and the line $x+y=1$.



$$\begin{array}{c} x \\ \hline 0 \leq x \leq 1-y \end{array}$$

This region is type I as well as type II.

D can be considered as type II region.

Left curve: $x = 0 \leftarrow r(y)$.

Right curve: $x = 1 - y \leftarrow s(y)$.

$$D = \{ (x, y) \in \mathbb{R}^2 / 0 \leq x \leq 1 - y, 0 \leq y \leq 1 \}.$$

$$\iint_D e^{x+y} dx dy = \int_{y=0}^1 \left[\int_{x=0}^{1-y} e^{x+y} dx \right] dy$$

$$= \int_{y=0}^1 \left[\left\{ e^{x+y} \right\}_{x=0}^{1-y} \right] dy$$

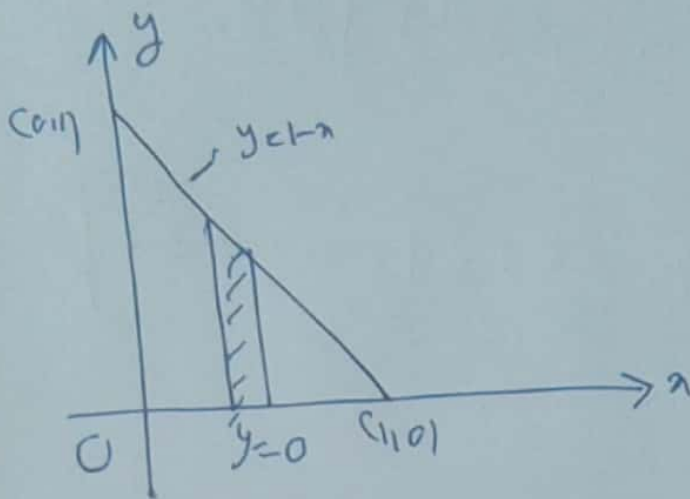
$$= \int_{y=0}^1 (e^{1-y+y} - e^{0+y}) dy$$

$$= \int_{y=0}^1 (e - e^y) dy$$

$$= \left[ey - e^y \right]_{y=0}^1$$

$$= 1 - 0$$

Evaluate $\iint_D e^{xy} dx dy$ using type I region.



$$\iint_D e^{xy} dy dx = \int_{x=0}^1 \left[\int_{y=0}^{1-x} e^{xy} dy \right] dx$$

$$= \int_0^1 \left[e^{xy} \right]_{y=0}^{1-x} dx$$

$$= \int_0^1 (e^{x+1-x} - e^{x \cdot 0}) dx$$

$$= \int_0^1 (e - e^x) dx$$

$$= \left[xe - e^x \right]_{x=0}^1$$

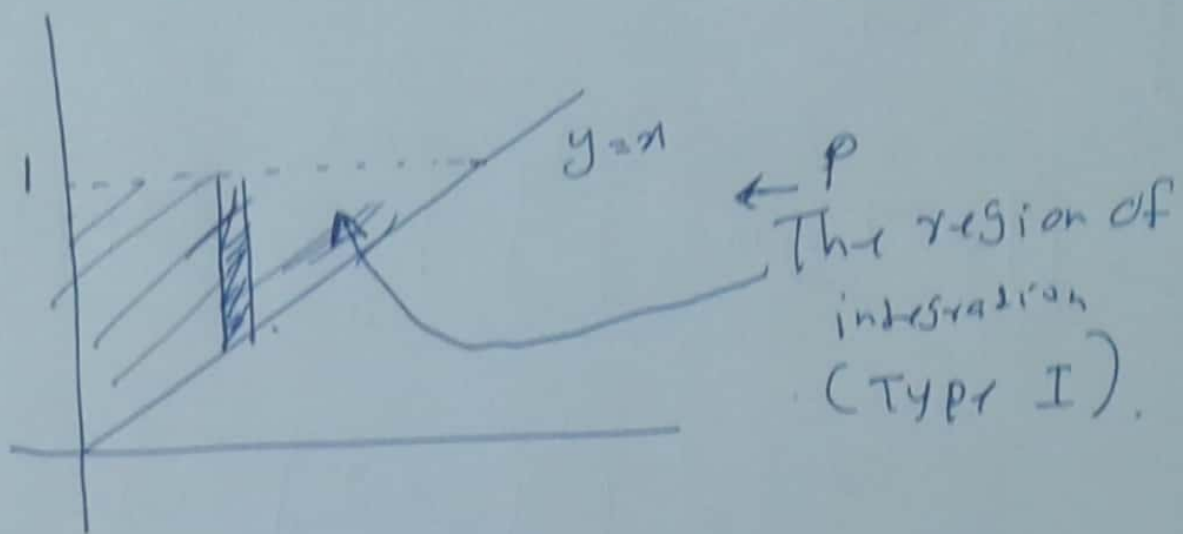
$$= (e - e) - (0 - 1)$$

$$= 1$$

Example.

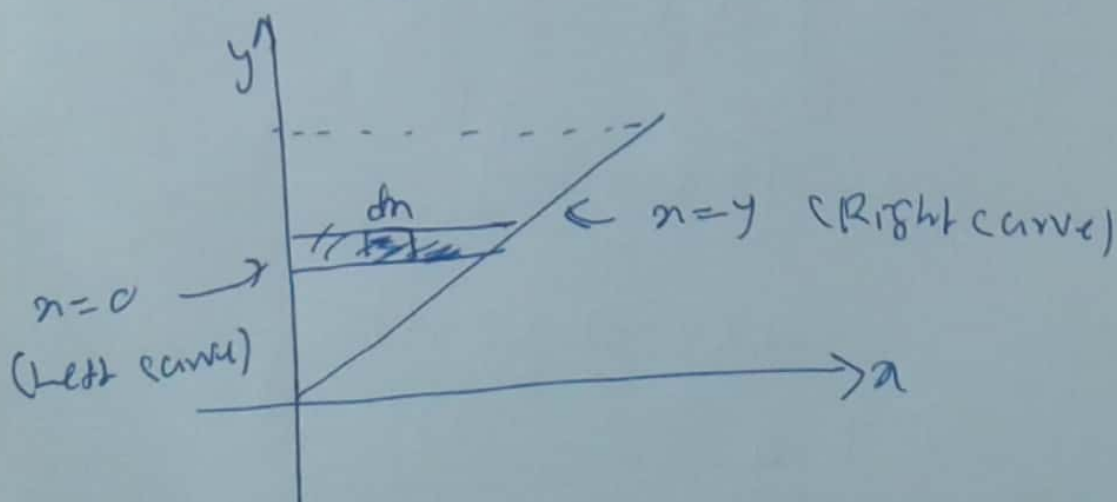
Evaluate $\int_0^1 \left[\int_x^1 \sin(y^2) dy \right] dx.$

$$D = \left\{ (x, y) \in \mathbb{R}^2 \mid \text{such that } x \leq y \leq 1, 0 \leq x \leq 1 \right\}.$$



$$\int_{x=0}^1 \left[\int_{y=x}^1 \sin(y^2) dy \right] dx = \iint_D \sin(y^2) dx dy \quad \text{--- (1)}$$

Next consider D as type II region.



$$\int_0^1 \int_0^y \sin(y^2) dx dy$$

$$= \int_{y=0}^1 \left[\int_{x=0}^y \sin(y^2) dx \right] dy$$

$$= \int_{y=0}^1 \left[x \sin(y^2) \right]_{x=0}^y dy$$

$$= \int_{y=0}^1 (y \sin(y^2) - 0) dy$$

$$= \int_{y=0}^1 y \sin(y^2)$$

$$= \left[\frac{-\cos(y^2)}{2} \right]_{y=0}^1$$

$$= -\frac{1}{2} (\cos(1^2) - \cos(0^2))$$

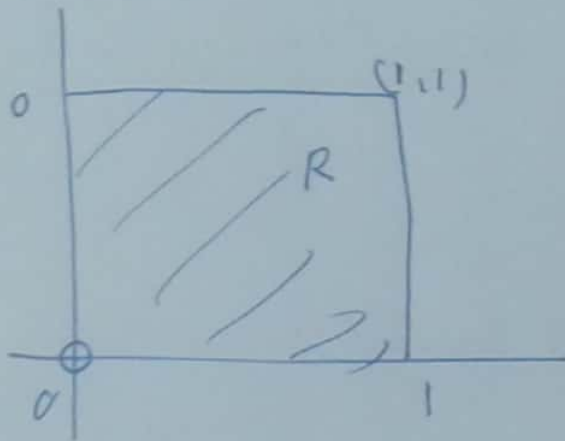
$$= \frac{1}{2} (1 - \cos(1))$$

Example

Show that $\int_0^1 \int_0^1 \frac{x-y}{(x+y)^3} dx dy = -1/2$ and

$$\int_0^1 \int_0^1 \frac{x-y}{(x+y)^3} dy dx = 1/2.$$

$$D = \{ (x,y) \in \mathbb{R}^2 / 0 \leq x \leq 1, 0 \leq y \leq 1 \} \setminus \{ (0,0) \}$$



$$I_1 = \int_0^1 \int_0^1 \frac{x-y}{(x+y)^3} dx dy$$

$$= \int_0^1 \int_0^1 \left(\frac{1}{(x+y)^2} - \frac{2y}{(x+y)^3} \right) dx dy$$

$$= \int_0^1 \left[\int_0^1 \left(\frac{1}{(x+y)^2} - \frac{2y}{(x+y)^3} \right) dx \right] dy$$

$$= \int_0^1 \left[\frac{-1}{(x+y)} - \frac{2y}{-2(x+y)^2} \right]_{x=0}^1 dy$$

$$= \int_0^1 \left[\frac{-1}{1+y} + \frac{2}{(1+y)^2} - 0 \right] dy$$

$$= \int_0^1 \frac{-1}{(1+y)^2} dy$$

$$= - \int_0^1 (1+y)^{-2} dy$$

$$= - \left[\frac{(1+y)^{-1}}{-1} \right]_{y=0}^1$$

$$= -\frac{1}{2}$$

$$I_2 = \int_0^1 \int_0^1 \left(\frac{2x}{(x+y)^3} - \frac{1}{(x+y)^2} \right) dy dx$$

$$= \int_0^1 \int_0^1 \left(\frac{2x}{(x+y)^3} - \frac{1}{(x+y)^2} \right) dy dx$$

$$I_2 = \int_0^1 \int_0^1 \left(\frac{2x}{(x-y)^3} - \frac{1}{(x-y)^2} \right) dy dx$$

$$= \int_0^1 \left[\frac{2x \cdot \frac{1}{-2(x-y)^2} + \frac{1}{(x-y)}}{y=0} \right]_{y=0}^1 dx$$

$$= \int_0^1 \left[\frac{-x}{(x-y)^2} + \frac{1}{(x-y)} \right]_{y=0}^1 dx$$

$$= \int_0^1 \left[\frac{1}{x+1} - \frac{x}{(x+1)^2} \right] dx$$

$$= \int_0^1 \frac{1}{(x+1)^2} dx$$

$$= \left[\frac{(x+1)^{-1}}{-1} \right]_{x=0}^1$$

$$= - \left[\frac{1}{2} - \frac{1}{1} \right]$$

$$= \frac{1}{2}$$

Example

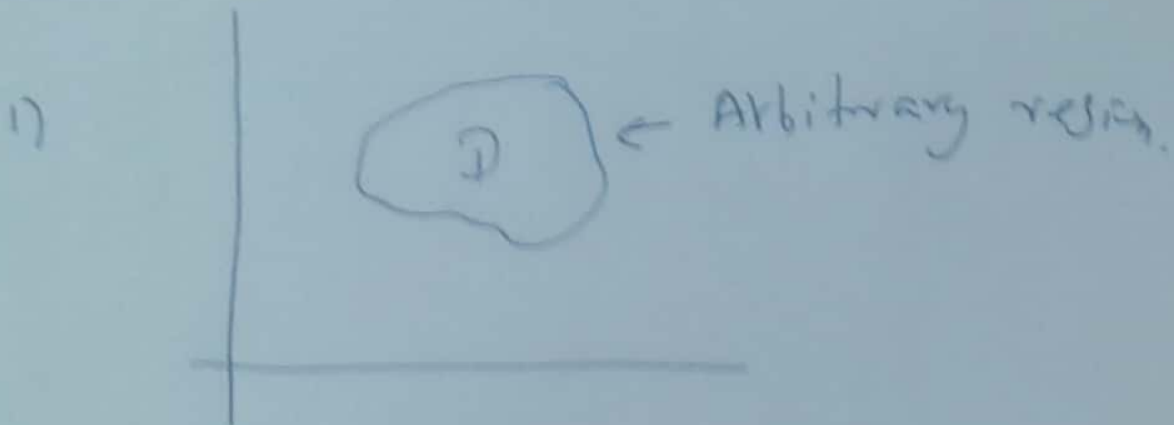
Note that:

According to Theorem 1,

$$f(x,y) = \frac{x-y}{(x-y)^2} \text{ cannot be integrated over}$$

$$R = \left\{ (x,y) / 0 \leq x \leq 1, 0 \leq y \leq 1 \right\}.$$

Note:



The area of the region $D = \iint_D 1 \cdot dA.$

2) If $f(x,y) = g(x)g(y)$, then

$$\int_a^b \int_c^d f(x,y) dy dx = \left[\int_a^b g(x) dx \right] \left[\int_c^d h(y) dy \right]$$

Example

A rectangular sheet lies in the region

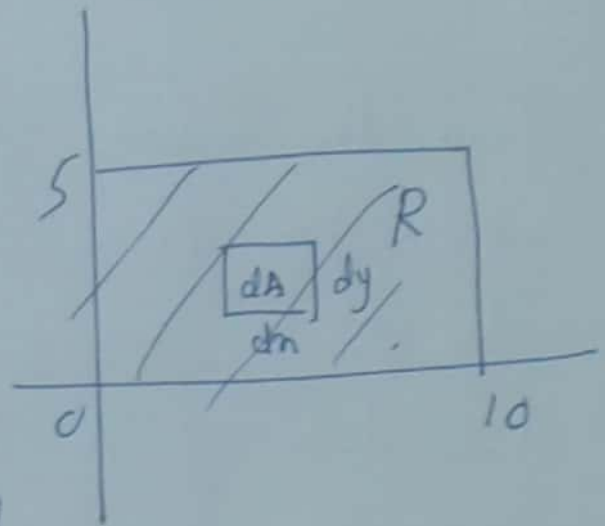
$$\therefore 0 \leq x \leq 10, 0 \leq y \leq 5.$$

Find the total mass of the sheet if the mass per unit area of the sheet is given by $m(x, y) = xy$.

$$m = m(x, y) = xy.$$

$$dA = dx dy$$

$$\begin{aligned} \therefore \text{The mass of } dA &= dAm \\ &= m dA \\ &= xy dx dy. \end{aligned}$$



$$\begin{aligned} \therefore \text{Total mass } M &= \iint_R m(x, y) dA \\ &= \int_{y=0}^5 \int_{x=0}^{10} xy dx dy \end{aligned}$$

$$= \left[\int_{y=0}^5 y \, dy \right] \left[\int_{x=0}^{10} x \, dx \right]$$

$$= \left[\frac{y^2}{2} \right]_{y=0}^5 \left[\frac{x^2}{2} \right]_{x=0}^{10}$$

$$= \frac{1}{2} (25) \cdot \frac{1}{2} (100)$$

$$= 625 \text{ units.}$$

Hom

1. Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region D in the xy -plane bounded by the line $y = 2x$ and the parabola $y = x^2$. [Ans: $\frac{216}{35}$]
2. Find the volume of the solid S' that is bounded ~~by the~~ by the elliptic paraboloid $x^2 + y^2 + x^2 + 2y^2 + z = 16$ the planes $x = 2$ and $y = 2$ and the coordinate planes [Ans: 48]